## 1 Workload and dependencies

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Alternating | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| Seed 42 prefix | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |

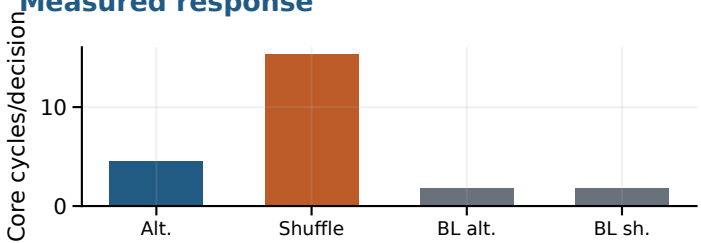
## 2 Controlled comparison

Identical 64 KiB addresses; constants / alternating / balanced shuffled; two seeds.

## 3 Timed batch and normalization

I=1,048,576 decisions; ticks/decision = elapsed TSC / I; branch retained in assembly.

## 4 Measured response




## 5 Inference and control

Shuffled outcomes increase native misses and core-cycle cost. The same outcome change barely affects branchless arithmetic, which helps distinguish prediction from the cost of reading the array. No single-miss penalty is inferred.